ImageSNN Evaluation Report

Generated: 13 August 2026 at 20:48
Images Tested: 700
Model Version: 1.0.0
Generation Manifest: generation_manifest_20260715_095824.json

Evaluation Summary

Metric	Value
Accuracy	0.7729
Precision	0.8848
Recall	0.8450
F1 Score	0.8645
AUC	0.6344
MSE	0.2200

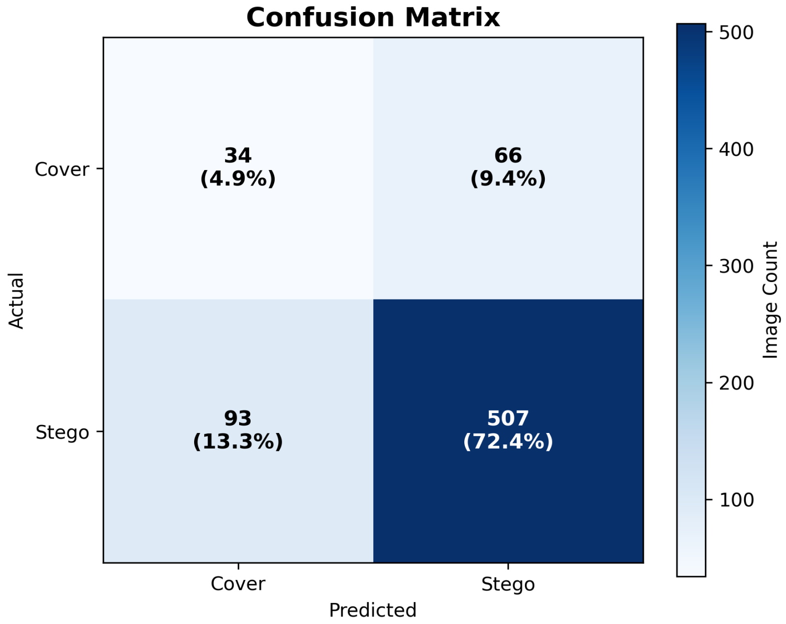
Key Findings

- Model accuracy achieved 77.29%.
- Recall reached 84.50%, indicating successful detection of most stego images.
- Error rate was 22.71% across analysed images.
- Mean inference latency was 336.89 ms with a throughput of 2.97 images/s.

Validation Summary

Validation reporting will be added in a future release.

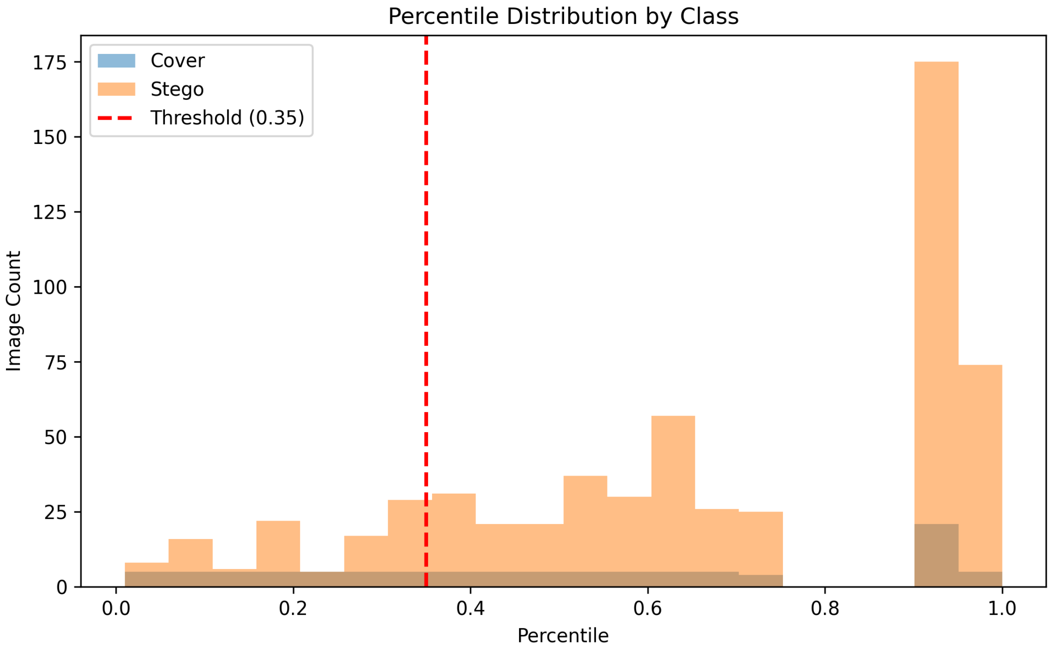
Confusion Matrix



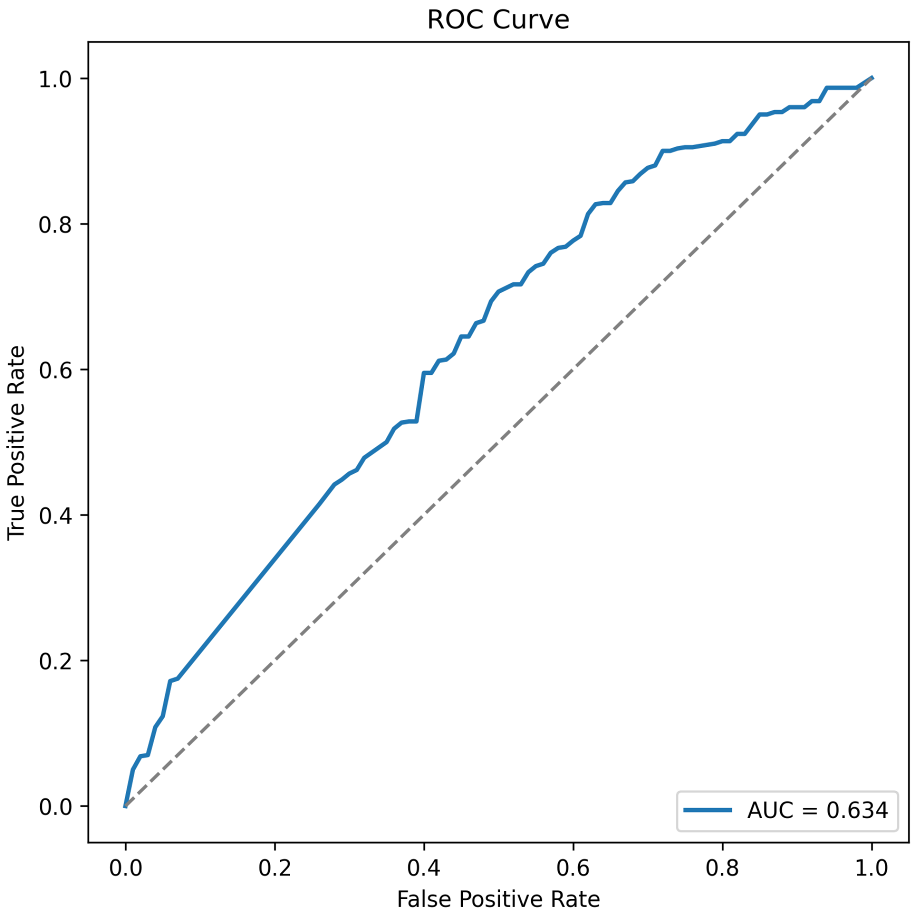
Summary
True Positives: 507
True Negatives: 34
False Positives: 66
False Negatives: 93

The model correctly classified 541 images and misclassified 159 images across the evaluation dataset.

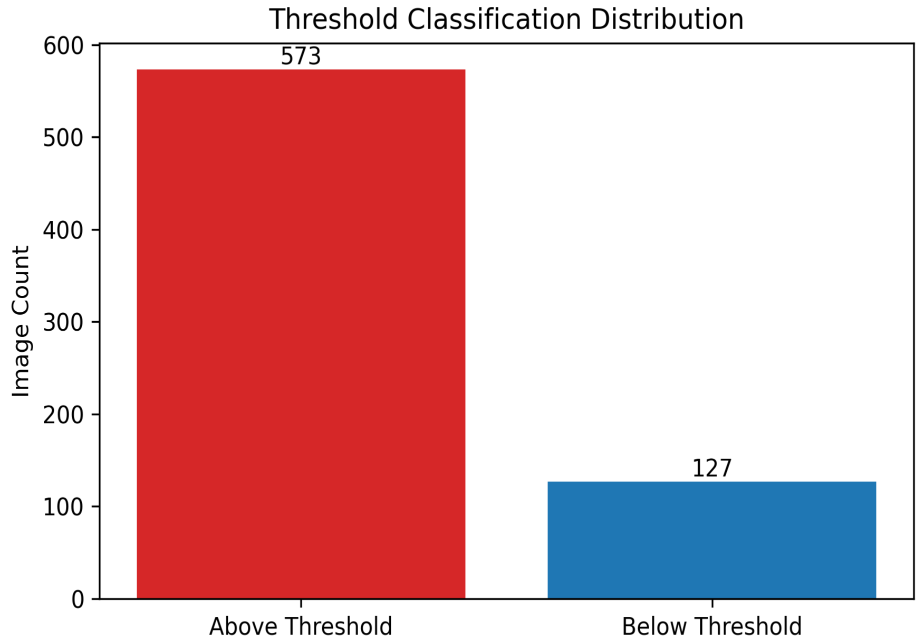
Probability Distribution



ROC Curve


ROC Curve
1.0
0.8
0.6
0.4
0.2
0.0
True Positive Rate
AUC = 0.634
0.0
0.2
0.4
0.6
0.8
1.0
False Positive Rate

Threshold Distribution



Summary
Above Threshold: 573
Below Threshold: 127
Threshold Proportion: 0.82%

0.82% of predictions exceeded the configured decision threshold.

Classification Outcomes

Mean Difference: 0.1389
False Positives: 66
False Negatives: 93
Above Threshold: 573
Below Threshold: 127
Threshold Proportion: 81.86%

Error Analysis Summary

Classification Results

Metric	Value
Pairs Analysed	600
True Positives	507
False Positives	66
True Negatives	34
False Negatives	93
Total Errors	159

Performance Metrics

Metric	Value
TPR (Recall)	84.50%
TNR (Specificity)	34.00%
FPR	66.00%
FNR	15.50%
Error Rate	22.71%

Dataset Quality

Folder	Avg PSNR	Avg SSIM
Stego Encrypted		
Large	56.2087	0.998619
Medium	61.2381	0.999562
Small	66.5664	0.999871
Stego Plain		
Large	56.1952	0.998616
Medium	61.2893	0.999568
Small	66.5479	0.999870

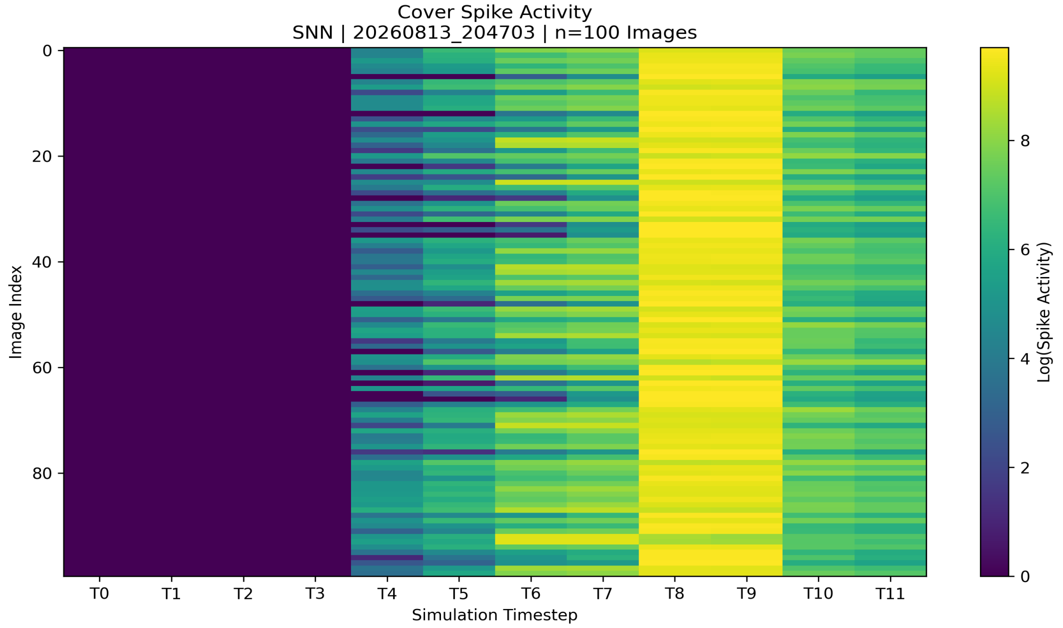
PSNR and SSIM metrics were used to assess visual similarity between cover and stego images. Higher PSNR and SSIM values indicate lower visual distortion and greater similarity to the original cover images.

Average PSNR and SSIM values are presented for reporting purposes. Detailed quality statistics, including minimum and maximum measurements, are provided within the accompanying workbook export.

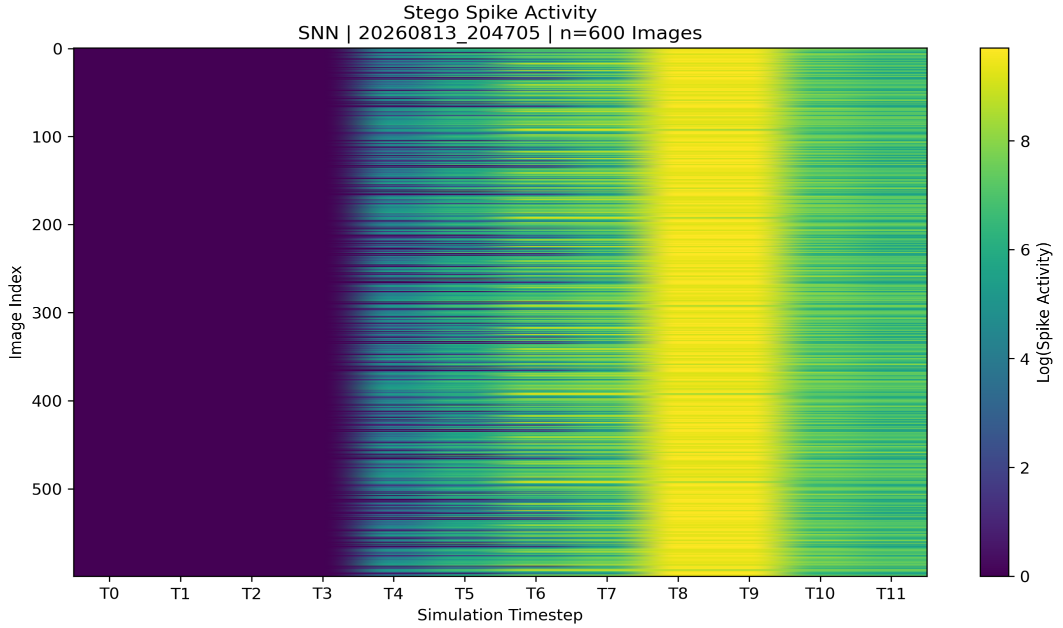
SNN Decision Analysis

Spike activation visualisations were generated to examine temporal firing behaviour within the hidden LIF layer of the ImageSNN architecture.

Cover Spike Activation Map



Stego Spike Activation Map



Generated: 13 August 2026 at 20:48

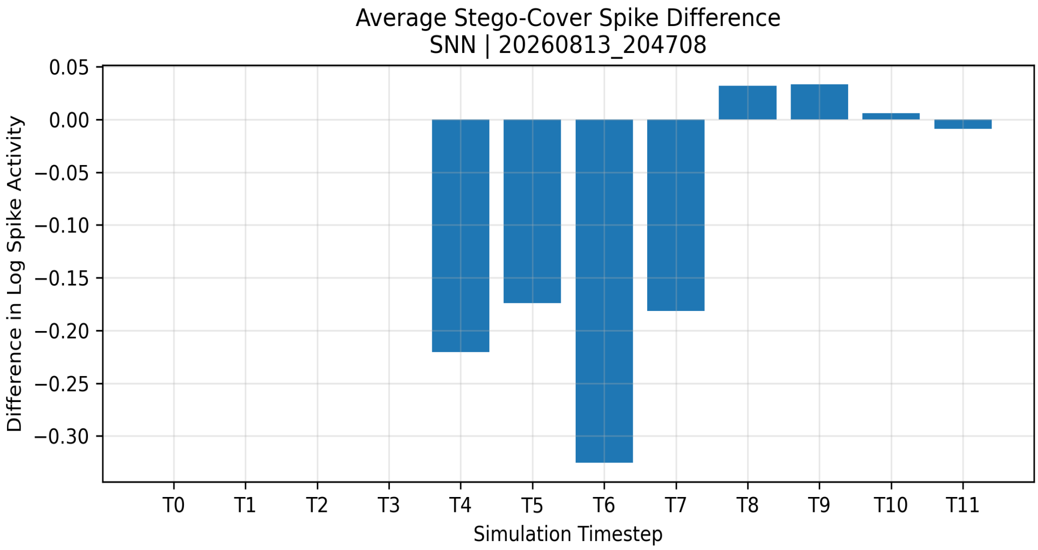
Model: ImageSNN v1.0.0

Research use only - not definitive forensic evidence.

Key Findings

Mean Spike Difference: -0.0700
Peak Difference Timestep: T6
Maximum Absolute Difference: -0.3256
Largest Positive Shift: 0.0333
Largest Negative Shift: -0.3256

Spike Difference Map



Spike Activity Analysis

The greatest spike-activity difference was observed at T6, where an absolute difference of -0.3256 was recorded. Spike activity was concentrated within the later simulation timesteps, suggesting that temporal firing behaviour diverged most strongly after membrane potentials had accumulated. The spike difference map highlights timesteps that contributed most strongly to classification differences between cover and stego images.

Performance Analysis

Mean Latency: 336.89 ms
Median Latency: 300.86 ms
Minimum Latency: 204.12 ms
Maximum Latency: 3337.85 ms
Images Processed: 700
Total Inference Time: 235.82 s
Throughput: 2.97 images/s

Detection Sensitivity Analysis

Payload Size Analysis

Category	Accuracy	Precision	Recall	F1	TNR	FPR
Small	71.50%	100.00%	71.50%	83.38%	0.00%	0.00%
Medium	83.00%	100.00%	83.00%	90.71%	0.00%	0.00%
Large	99.00%	100.00%	99.00%	99.50%	0.00%	0.00%

Payload Type Analysis

Category	Accuracy	Precision	Recall	F1	TNR	FPR
Plain	84.67%	100.00%	84.67%	91.70%	0.00%	0.00%
Encrypted	84.33%	100.00%	84.33%	91.50%	0.00%	0.00%

Payload Category Analysis

Category	Accuracy	Precision	Recall	F1	TNR	FPR
Encrypted Large	98.00%	100.00%	98.00%	98.99%	0.00%	0.00%
Encrypted Medium	83.00%	100.00%	83.00%	90.71%	0.00%	0.00%
Encrypted Small	72.00%	100.00%	72.00%	83.72%	0.00%	0.00%
Plain Large	100.00%	100.00%	100.00%	100.00%	0.00%	0.00%
Plain Medium	83.00%	100.00%	83.00%	90.71%	0.00%	0.00%
Plain Small	71.00%	100.00%	71.00%	83.04%	0.00%	0.00%

Research Interpretation

Payload Size

Large payloads achieved the strongest performance (F1: 99.50%), while Small payloads produced the weakest results (F1: 83.38%). This suggests that detection performance improves as payload volume increases.

Payload Type

Payload encryption had negligible impact on performance, with plain and encrypted payloads achieving comparable F1 scores (91.70% and 91.50%, respectively).

Research Insight

The observed performance trend suggests that ImageSNN is responding primarily to embedding artefacts rather than payload content. This behaviour is consistent with findings obtained from ImageCNN despite the substantial architectural differences between the two models.

Note: AUC and MSE metrics are not reported for sensitivity-analysis subsets because these subsets contain only stego samples. Both metrics require a mixture of cover and stego images to provide meaningful comparative measurements. Consequently, the sensitivity analysis focuses on accuracy, precision, recall and F1 score, which remain valid indicators of detection performance within individual payload groups.

Limitations

This report provides an automated assessment based on the configured steganalysis model and evaluation parameters. Detection results represent probabilistic indicators rather than definitive proof of steganographic activity. False positives and false negatives may occur, particularly when processing previously unseen datasets, alternative embedding techniques, heavily compressed images, or images that differ significantly from the training data. Results should therefore be interpreted in conjunction with additional forensic analysis, supporting evidence, and professional judgement. Model performance is influenced by dataset characteristics, payload size, image format, calibration thresholds, and other experimental factors documented elsewhere in this report.